

DRY-RUN GUIDE | merger.py
Source: Dsa\sort\merger.py

CORE BEHAVIOR

Orders values using the comparisons, swaps, insertion, partition, or merges shown in this file.

REPRESENTATIVE INPUT

[4,1,3,2].

STEP TRACE

Each pass/partition places values nearer their final positions; completed sequence is nondecreasing.

EXPECTED OUTPUT / RETURN

1 2 3 4 (check source for exact formatting).

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Dsa\sort\merger.py; line numbers refer to the original file. No source was modified.

```
001 | def merge_sort(arr):
002 |     if len(arr) <= 1:
003 |         return arr
004 |
005 |     mid = len(arr) // 2
006 |     left = merge_sort(arr[:mid])
007 |     right = merge_sort(arr[mid:])
008 |
009 |     return merge(left, right)
010 |
011 | def merge(left, right):
012 |     result = []
013 |     i = j = 0
014 |     while i < len(left) and j < len(right):
015 |         if left[i] <= right[j]:
016 |             result.append(left[i])
017 |             i += 1
018 |         else:
019 |             result.append(right[j])
020 |             j += 1
021 |     result.extend(left[i:])
022 |     result.extend(right[j:])
023 |     return result
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.